

EXP 5 (5.1)

```
clc; clear; close all;
%% --- Input Parameters ---
fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order

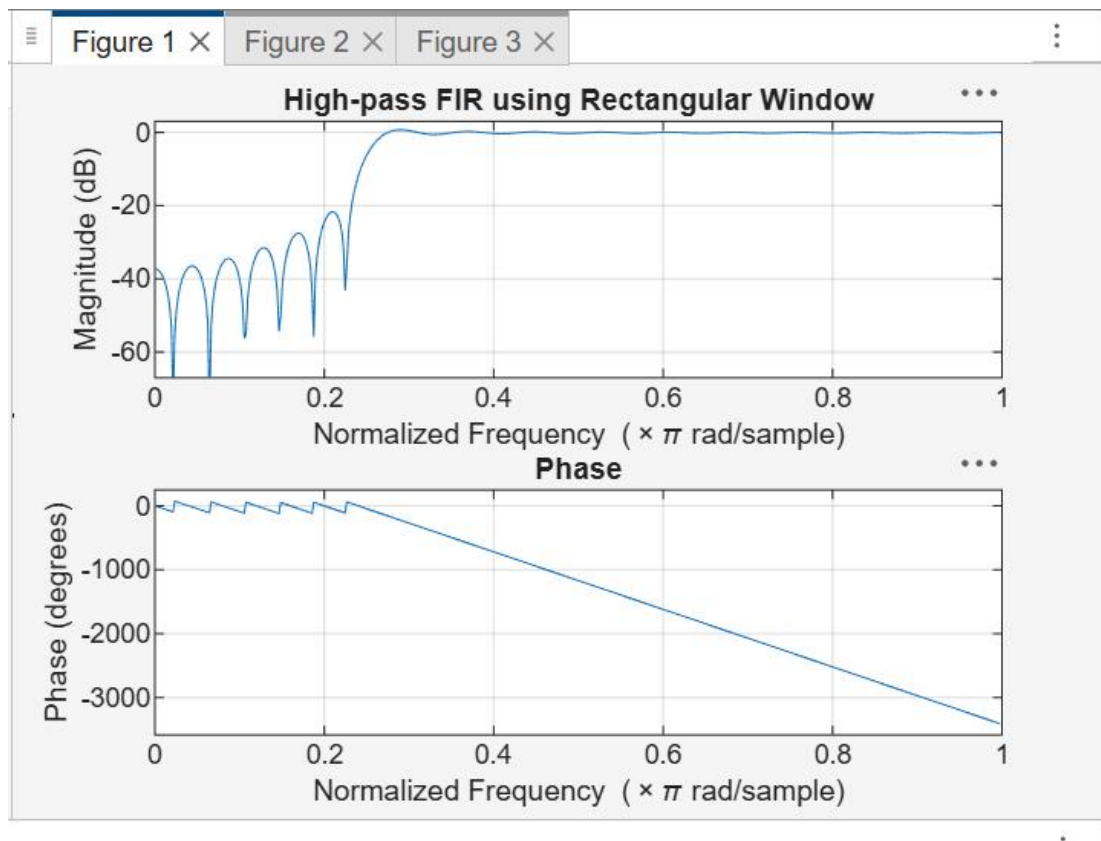
wc = fp/(fs/2); % Normalized cutoff (0-1)

%% --- 1. FIR High Pass using Rectangular Window ---
b_rect = fir1(N, wc, 'high', rectwin(N+1));
disp('--- High-Pass Coefficients (Rectangular Window) ---');
disp(b_rect');

%% --- 2. FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));
disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm');

%% --- 3. FIR High Pass using Hanning Window ---
b_hann = fir1(N, wc, 'high', hanning(N+1));
disp('--- High-Pass Coefficients (Hanning Window) ---');
disp(b_hann');

%% --- Frequency Response Plots ---
figure;
freqz(b_rect,1);
title('High-pass FIR using Rectangular Window');
figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
figure;
freqz(b_hann,1);
title('High-pass FIR using Hanning Window');
```



--- High-Pass Coefficients (Rectangular Window) ---

-0.0089

0

0.0097

0.0143

0.0106

0

-0.0117

-0.0175

-0.0131

0

0.0148

0.0225

0.0171

0

-0.0202

-0.0314

-0.0247

0

0.0318

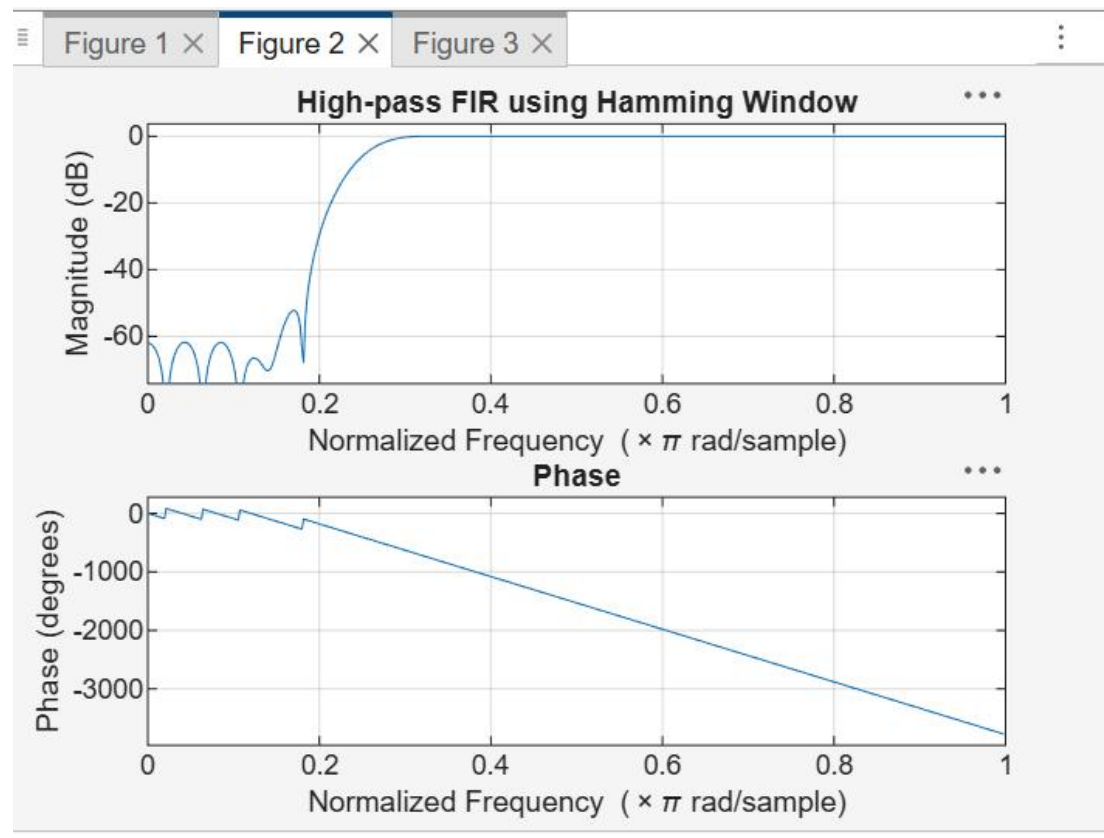
0.0524

0.0445

EXP 5 (5.2)

```
clc; clear; close all;
%% --- Input Parameters ---
fp = 1000; % High-pass cutoff frequency (Hz)
fs = 8000; % Sampling frequency (Hz)
N = 50; % Filter order
wc = fp/(fs/2); % Normalized cutoff (0-1)
%% --- FIR High Pass using Hamming Window ---
b_hamm = fir1(N, wc, 'high', hamming(N+1));

disp('--- High-Pass Coefficients (Hamming Window) ---');
disp(b_hamm);
%% --- Frequency Response Plots ---
figure;
freqz(b_hamm,1);
title('High-pass FIR using Hamming Window');
title('High-pass FIR using Hanning Window');
```



```

--- High-Pass Coefficients (Hamming Window) ---
-0.0007
    0
    0.0009
    0.0016
    0.0015
    0
-0.0024
-0.0044
-0.0039
    0
    0.0060
    0.0103
    0.0088
    0
-0.0128
-0.0217
-0.0184
    0
    0.0268
    0.0464
    0.0410

```

EXP 5(5.3)

```
clc; clear; close all;
```

```
%% --- Input Parameters ---
```

```
fp = 1000; % High-pass cutoff frequency (Hz)
```

```
fs = 8000; % Sampling frequency (Hz)
```

```
N = 50; % Filter order
```

```
wc = fp/(fs/2); % Normalized cutoff (0-1)
```

```
%% --- FIR High Pass using Hanning Window ---
```

```
b_hann = fir1(N, wc, 'high', hanning(N+1));
```

```
disp('--- High-Pass Coefficients (Hanning Window) ---');
```

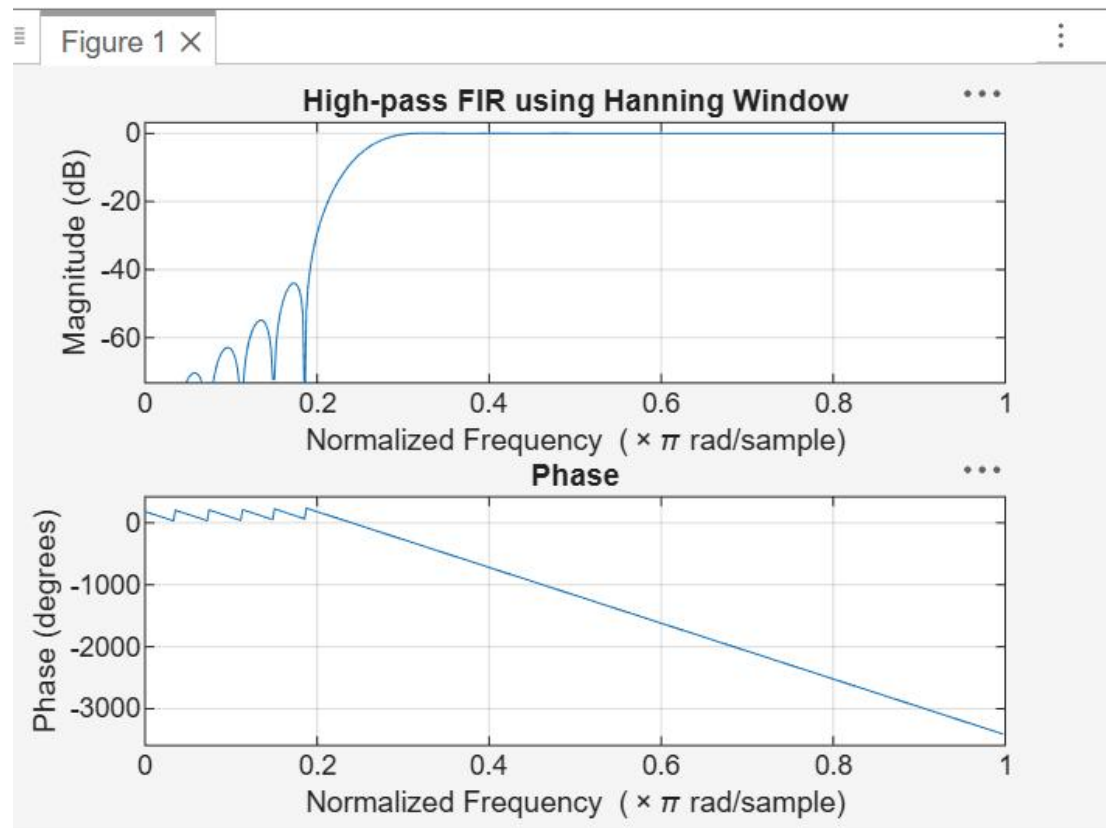
```
disp(b_hann');
```

```
%% --- Frequency Response Plots ---
```

```
figure;
```

```
freqz(b_hann,1);
```

```
title('High-pass FIR using Hanning Window');
```



--- High-Pass Coefficients (Hanning Window) ---

-0.0000

0

0.0003

0.0008

0.0009

0

-0.0020

-0.0038

-0.0035

0

0.0057

0.0100

0.0087

0

-0.0127

-0.0216

-0.0183

0

0.0267

0.0464

0.0410